The Decline of Traditional Retail

and the Rise of E-Commerce in the United States

A Data Visualization Analysis

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1. Introduction

1.1 Background

The American retail landscape has changed significantly over the past two decades. E-commerce started as a niche channel for books and electronics, but it has since grown into a major force that is reshaping consumer behavior, commercial real estate, and the retail job market across the United States. The growth of online shopping, driven by faster internet, smartphones, and platforms like Amazon, has changed how Americans shop and where retail jobs are created and lost.

The most visible sign of this shift has been the decline of physical retail. Department stores that once anchored shopping malls across the country have closed locations or filed for bankruptcy. Mall vacancy rates have hit historic highs in many areas. At the same time, fulfillment centers and logistics networks have expanded rapidly to handle the increase in online orders. This split between the fortunes of digital and physical retail is one of the most significant economic shifts in recent memory.

The COVID-19 pandemic accelerated trends that were already in motion. When stores closed in March 2020, consumers who had avoided online shopping were pushed to adopt it, and many kept those habits after stores reopened. E-commerce retained a higher share of total retail sales after the pandemic than it held before, suggesting the shift is structural and not just a temporary reaction.

This analysis uses publicly available data from the U.S. Census Bureau, the Federal Reserve Bank of St. Louis, and the U.S. Bureau of Labor Statistics to examine the scale and pace of this transformation across categories, workforce outcomes, and geography.

1.2 Research Questions

This analysis addresses two research questions:

Research Question 1: How has e-commerce grown as a share of total U.S. retail sales over time, and which product categories have experienced the greatest shift from in-store to online purchasing?

Research Question 2: How does the pace of e-commerce adoption vary across retail categories, and what does the trajectory of traditional brick-and-mortar sales suggest about where the shift is still ongoing?

2. Methodology

2.1 Data Sources

This analysis uses four data sources. Each one covers a different dimension of the retail story and is publicly available at no cost.

Source 1: U.S. E-Commerce Retail Sales (FRED)

The Federal Reserve Bank of St. Louis hosts the U.S. Census Bureau quarterly e-commerce retail sales series. It tracks total e-commerce sales in millions of dollars from Q4 1999 through Q3 2025 and provides the core time series for measuring how e-commerce has grown as a share of total retail.

URL: https://fred.stlouisfed.org/series/ECOMSA

Source 2: Quarterly E-Commerce Supplemental Tables (U.S. Census Bureau)

The Census Bureau publishes supplemental quarterly tables that break down e-commerce and total retail sales by major category. This dataset covers Q1 2018 through Q4 2024 and allows for the calculation of e-commerce penetration rates by category, meaning the share of each category's total sales that occurred online. Categories include clothing and general merchandise, furniture and electronics, motor vehicle parts, food and beverage, non-store retailers, and others.

URL: https://www.census.gov/retail/ecommerce/historic_releases.html

Source 3: Retail Employment by Sector (BLS via FRED)

The Bureau of Labor Statistics tracks employment across retail subsectors through its Current Employment Statistics program. Annual employment data for department stores, clothing stores, electronics stores, and non-store retailers were pulled from FRED for 2000 through 2024. Employment figures are indexed to 2000 (2000 = 100) so that relative changes across sectors of different sizes can be compared directly.

URL: https://fred.stlouisfed.org

Source 4: Monthly State Retail Sales (U.S. Census Bureau)

The Census Bureau's Monthly State Retail Sales program publishes year-over-year percentage changes in retail sales by state and retail subsector, covering 2019 through 2025. This dataset adds the geographic dimension to the analysis, covering all 50 states and the District of Columbia. This dataset covers physical retail only and does not include non-store retailers.

URL: https://www.census.gov/retail/state_retail_sales.html

2.2 Data Model

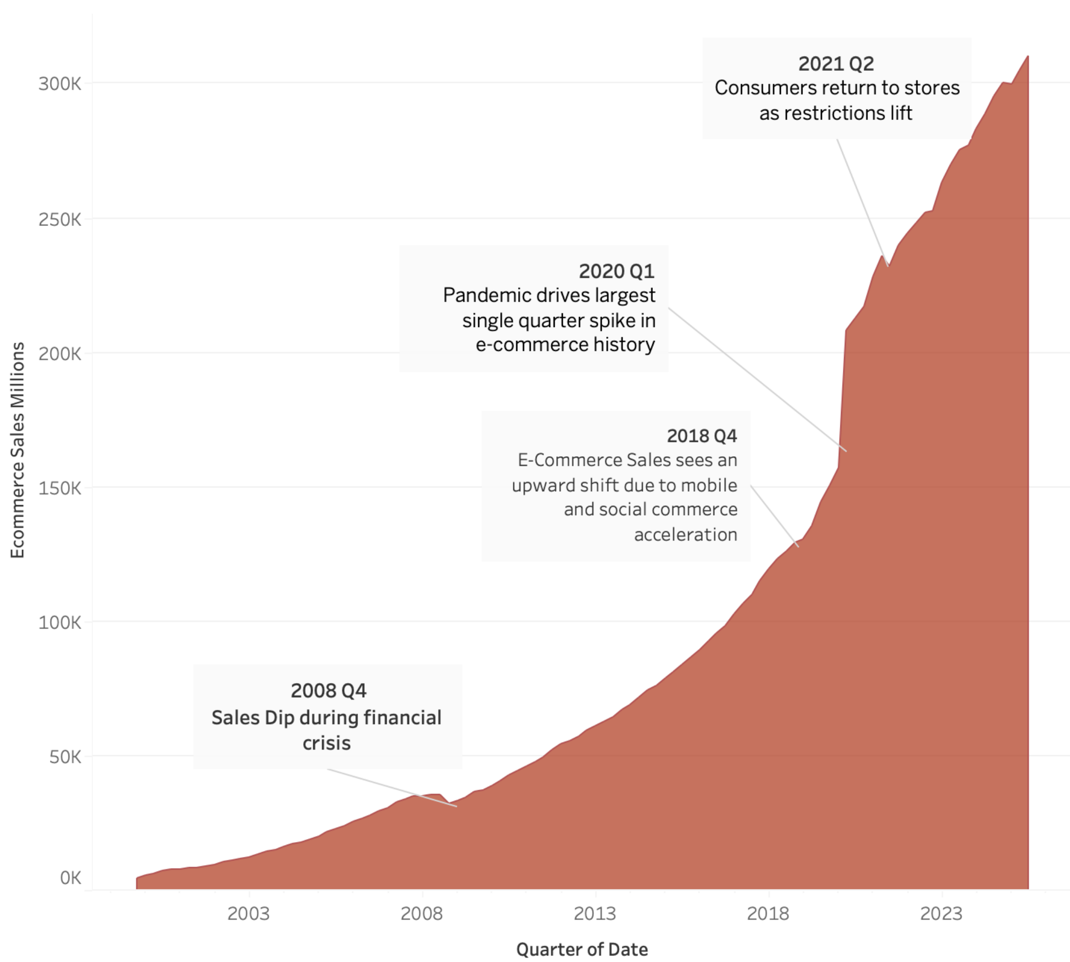
The four data sources were connected in Tableau using a star schema. A dimension table with retail category identifiers, category names, and category groupings (brick-and-mortar vs. online) sits at the center and links to each of the four fact tables through a shared Category ID field. All data preparation was done in Python before loading into Tableau, including reshaping files from wide to long format, standardizing category names, and calculating metrics like the employment index and e-commerce penetration rates.

3. Analysis

3.1 The Overall Growth of E-Commerce (Research Question 1)

The quarterly e-commerce sales trend chart shows U.S. online retail sales from Q4 1999 through 2025, growing from roughly $4.5 billion per quarter in 1999 to over $300 billion per quarter by 2025. Three inflection points stand out. The first is a slowdown during the 2008 to 2009 financial crisis when overall consumer spending dropped. The second is an acceleration in Q4 2018, likely driven by the rise of mobile shopping and social commerce. The third is the sharp spike in Q2 2020 when pandemic lockdowns closed physical stores and pushed consumers online in large numbers. A brief pullback in Q2 2021 followed as stores reopened and some consumers returned to in-person shopping. Sales never fell back to pre-pandemic levels, however, which points to a lasting behavioral shift.

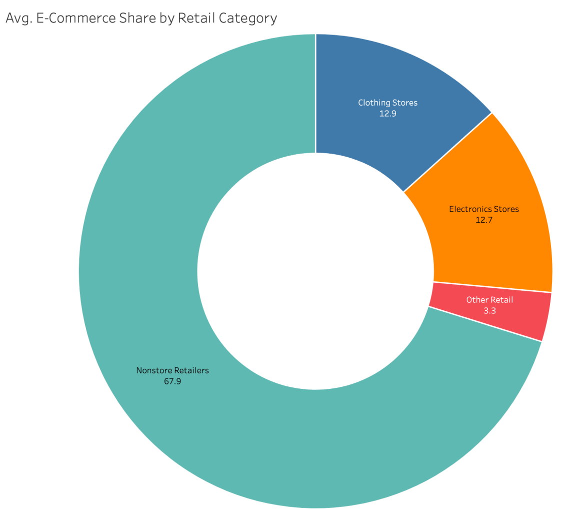
Quarterly U.S. E-Commerce Sales Over Time

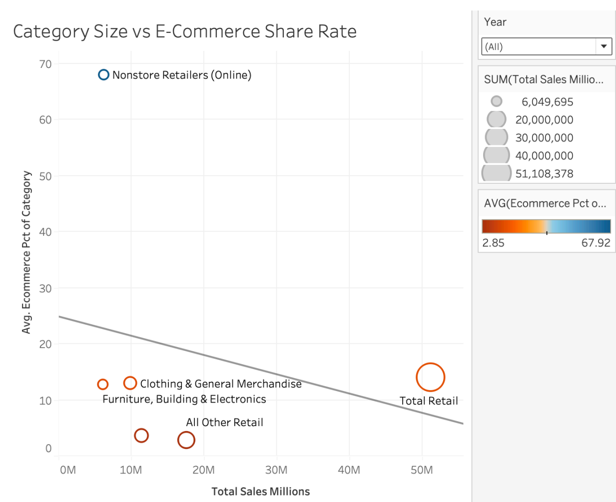


3.2 E-Commerce Penetration by Category (Research Question 1)

The radial chart and scatter plot show how much e-commerce adoption varies by retail category. Non-store retailers, whose primary sales channel is online, report penetration rates above 66%, making them a clear outlier. This is likely a reflection of their business model, not a transition away from physical stores. Among traditional retail categories, clothing and general merchandise have the highest average penetration at around 13%, followed by furniture and electronics at roughly 12%. Motor vehicles and parts have the lowest penetration, below 10%, which makes sense given that consumers tend to prefer evaluating high-value purchases in person.

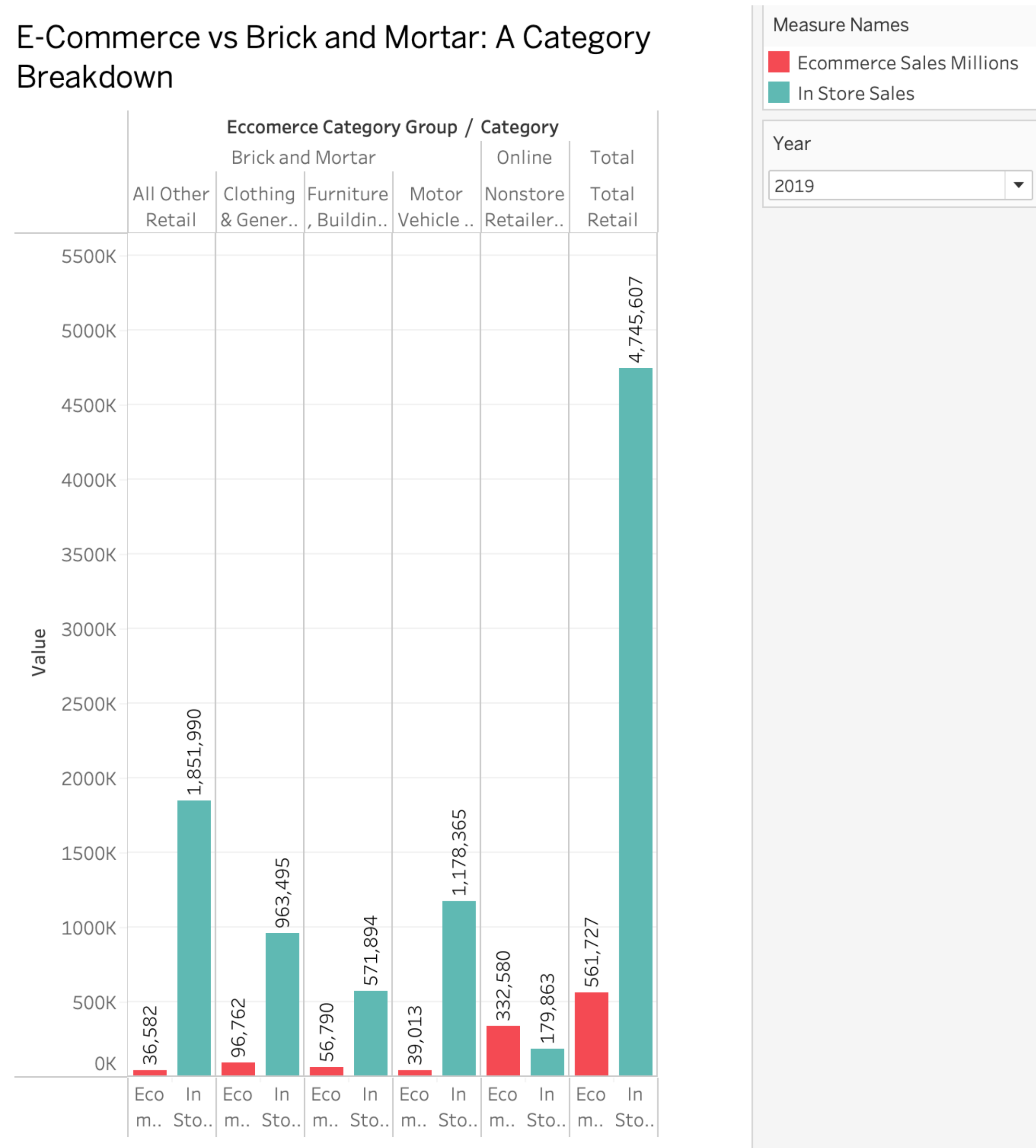
The scatter plot maps total category sales volume against average e-commerce penetration. The largest categories by sales are not always the most digitally adopted. Food and beverage, which falls under the ‘all other’ retail grouping, is one of the largest categories but has very low e-commerce penetration, meaning there is still a lot of room for further online growth there. Clothing, which is a moderately sized category, has a much higher penetration rate, driven by the ease of standardizing purchases and the success of online apparel platforms.





3.3 In-Store vs. Online Sales Comparison (Research Question 2)

The e-commerce versus brick-and-mortar bar chart lets users select a year using a parameter and see the balance between online and in-store sales for each category from 2018 to 2025. In-store sales still make up the majority for most categories, but the e-commerce share has grown every year. The drill-down by category group makes it easy to see the difference between brick-and-mortar retailers, where the in-store share is shrinking, and non-store retailers, where nearly all sales are already online. Clothing and general merchandise show the biggest shift toward online channels over the period.

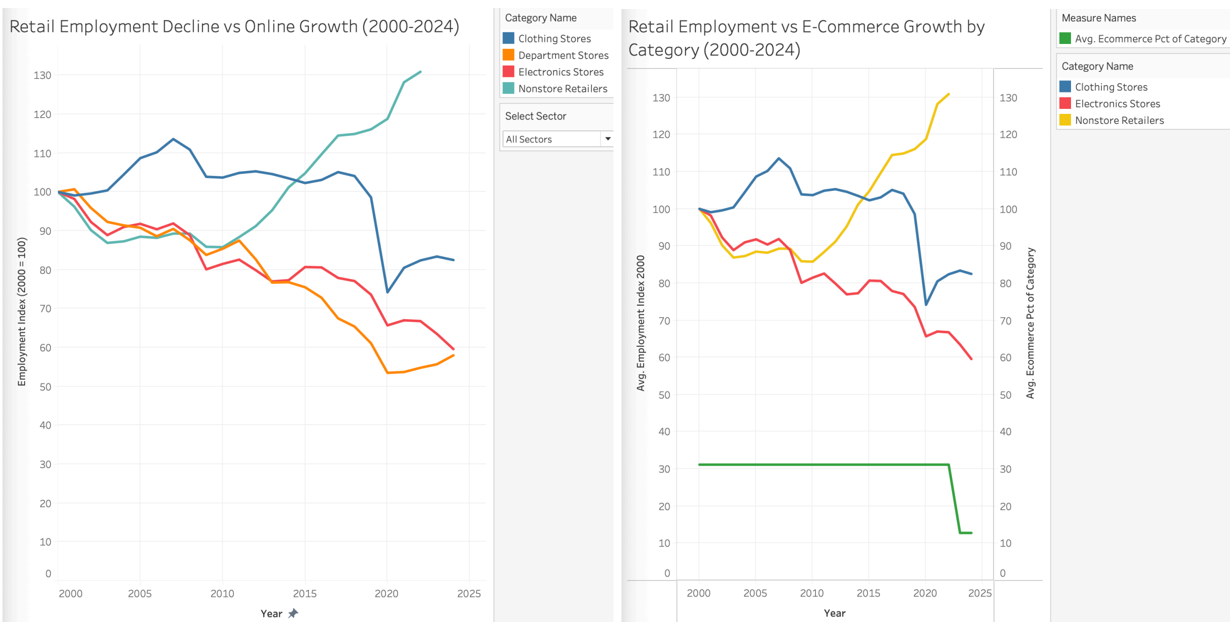


3.4 Workforce Implications (Research Question 2)

The retail employment index chart tracks employment for four sectors indexed to 100 in 2000. Electronics store employment has dropped the most, falling to about 60 by 2024, which is a loss of nearly 40% of the sector's workforce relative to its 2000 level. Department stores followed a similar path, declining from 2001 onward and accelerating after 2017. Clothing store employment stayed relatively flat through 2019 before dropping sharply in 2020 as online apparel purchasing spiked during the pandemic.

Non-store retailer employment went in the opposite direction, rising to about 130 by 2024, a 30% gain over the 2000 baseline. This growth reflects the expansion of fulfillment and logistics jobs that support e-commerce operations. These are largely warehouse and distribution roles rather than the customer-facing jobs that have been lost in physical stores, which means the retail workforce has not just shrunk but fundamentally changed in character.

The dual-axis chart overlaying the employment index with e-commerce penetration rates makes the connection between digital adoption and job loss visible. The sectors with the steepest employment declines, electronics and department stores, are also the ones with the highest e-commerce penetration rates.

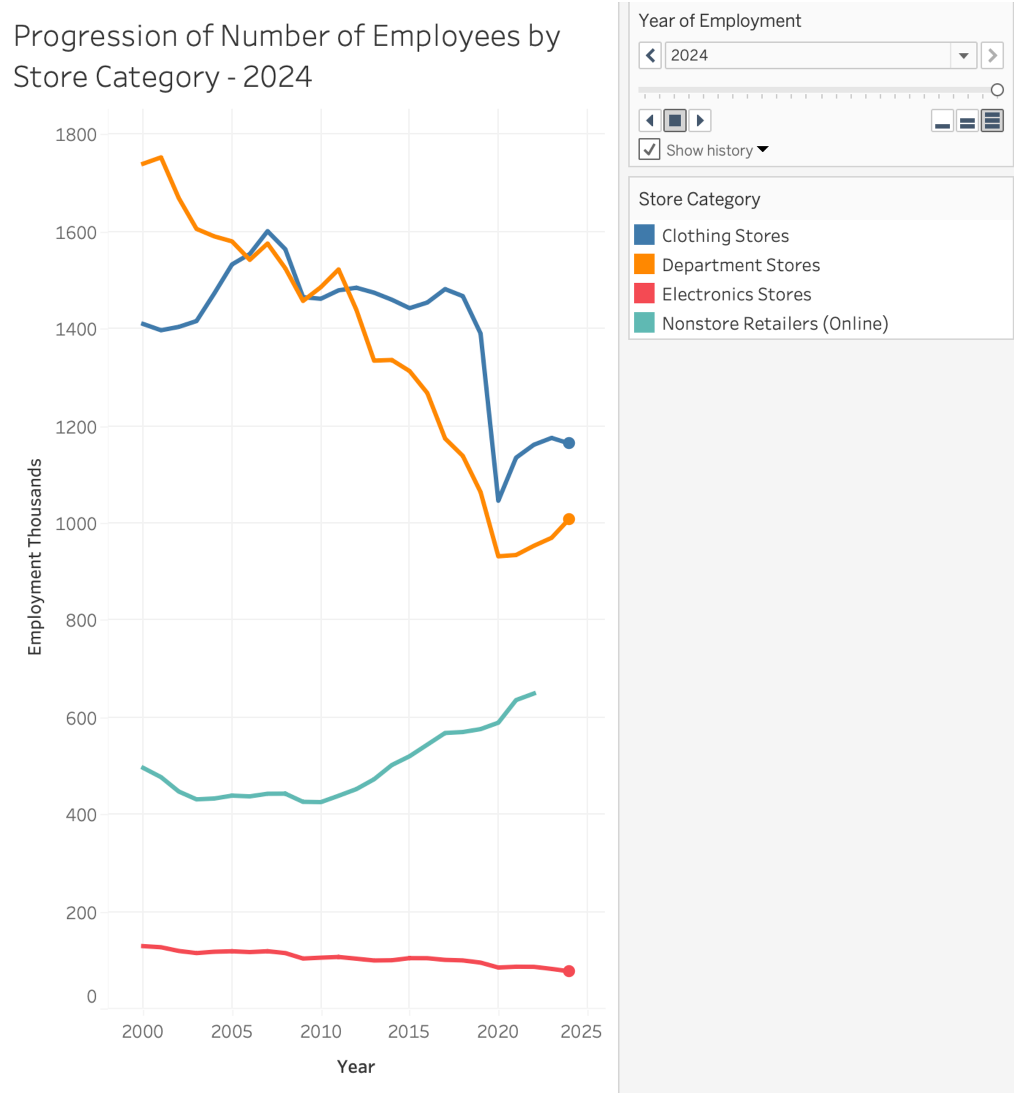


3.5 Progression of Retail Employment by Store Category

The animated chart steps through employment levels by store category from 2000 to 2024 one year at a time. This view makes the divergence between physical and online retail employment easy to follow as it unfolds over time.

In the early 2000s, all four categories start at similar indexed levels. As the years advance, a clear split develops. Department stores and electronics stores decline steadily, with the drop becoming steeper after 2015. Clothing stores hold relatively flat before falling sharply after 2019. Non-store retailers move in the opposite direction throughout, growing consistently and accelerating through the 2020s.

The animation also highlights the impact of specific events. The 2008 to 2009 financial crisis shows up as a broad dip across all sectors. The 2020 pandemic year shows a sharp contraction in physical retail employment for clothing and department stores, followed by a partial recovery that does not fully return to pre-pandemic levels. Non-store retailer employment accelerates sharply through 2020 and 2021 as online order volumes surged during lockdowns.

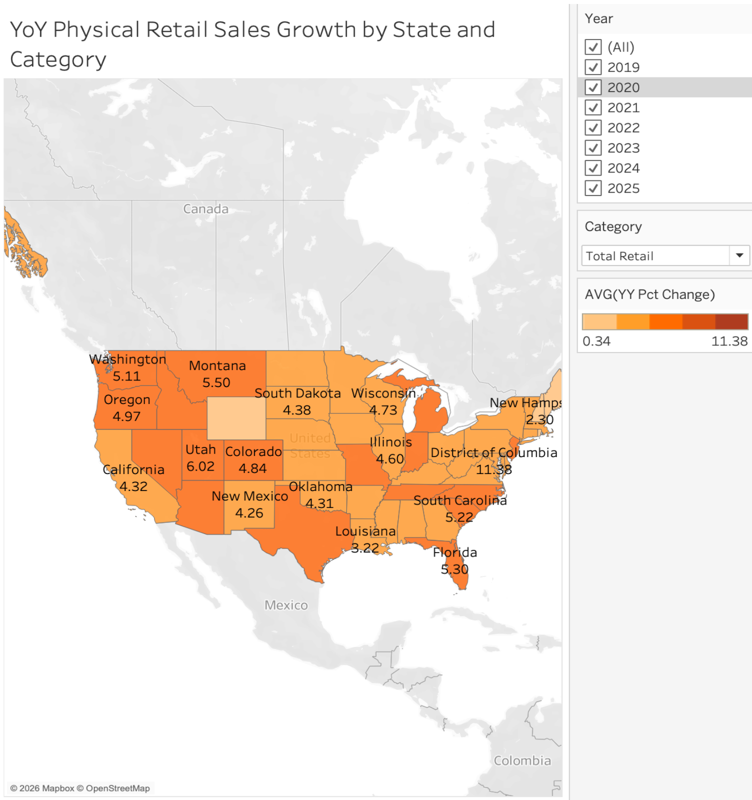


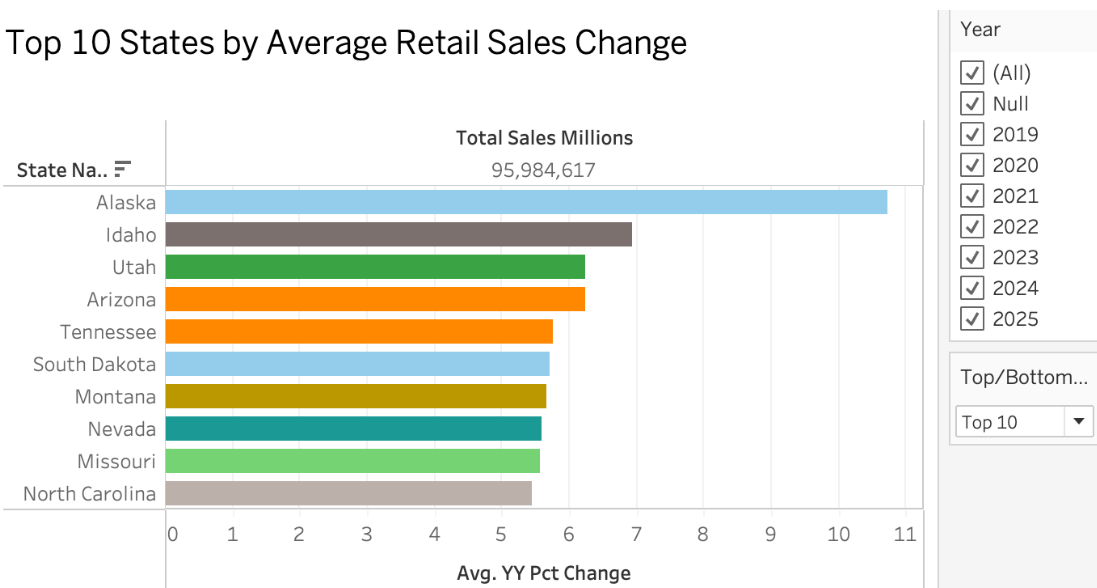
3.6 Geographic Variation (Research Question 2)

The choropleth map of year-over-year retail sales changes by state shows that physical retail performance varies significantly by geography. States in the South and Mountain West, including South Carolina, Florida, and South Dakota, show consistently positive year-over-year changes across most categories. This could suggest that population growth in these regions is keeping physical retail demand strong despite national headwinds. States in the Northeast and Mid-Atlantic, including New York and Connecticut, show more mixed results, with some categories registering persistent negative changes.

The category parameter on the map lets viewers filter by retail subsector. Clothing stores show the widest geographic variation of any category, with strong positive changes in some states and persistent declines in others. Food and beverage show relatively consistent positive changes across most states, which lines up with its low e-commerce penetration nationally.

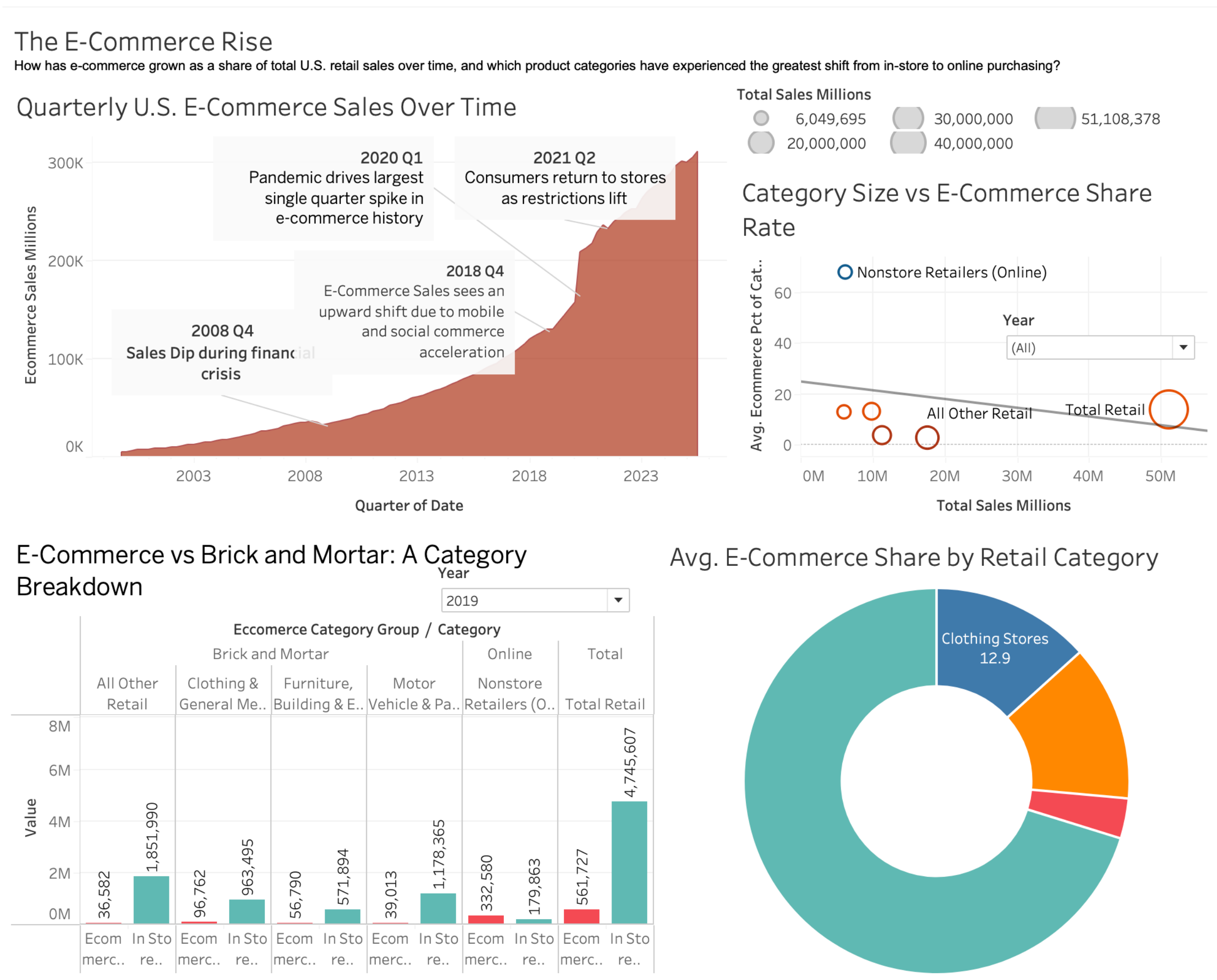
The top and bottom states chart confirms that the geographic variation is not random. States that consistently appear at the bottom could reflect slower-growing Northeastern markets with older populations and higher existing e-commerce adoption. The top performers are consistent with fast-growing Sun Belt states where new residents are supporting continued demand for physical retail. The total sales in millions discrete measure from the ecommerce table creates a viewpoint that zooms out to look at the bigger picture.





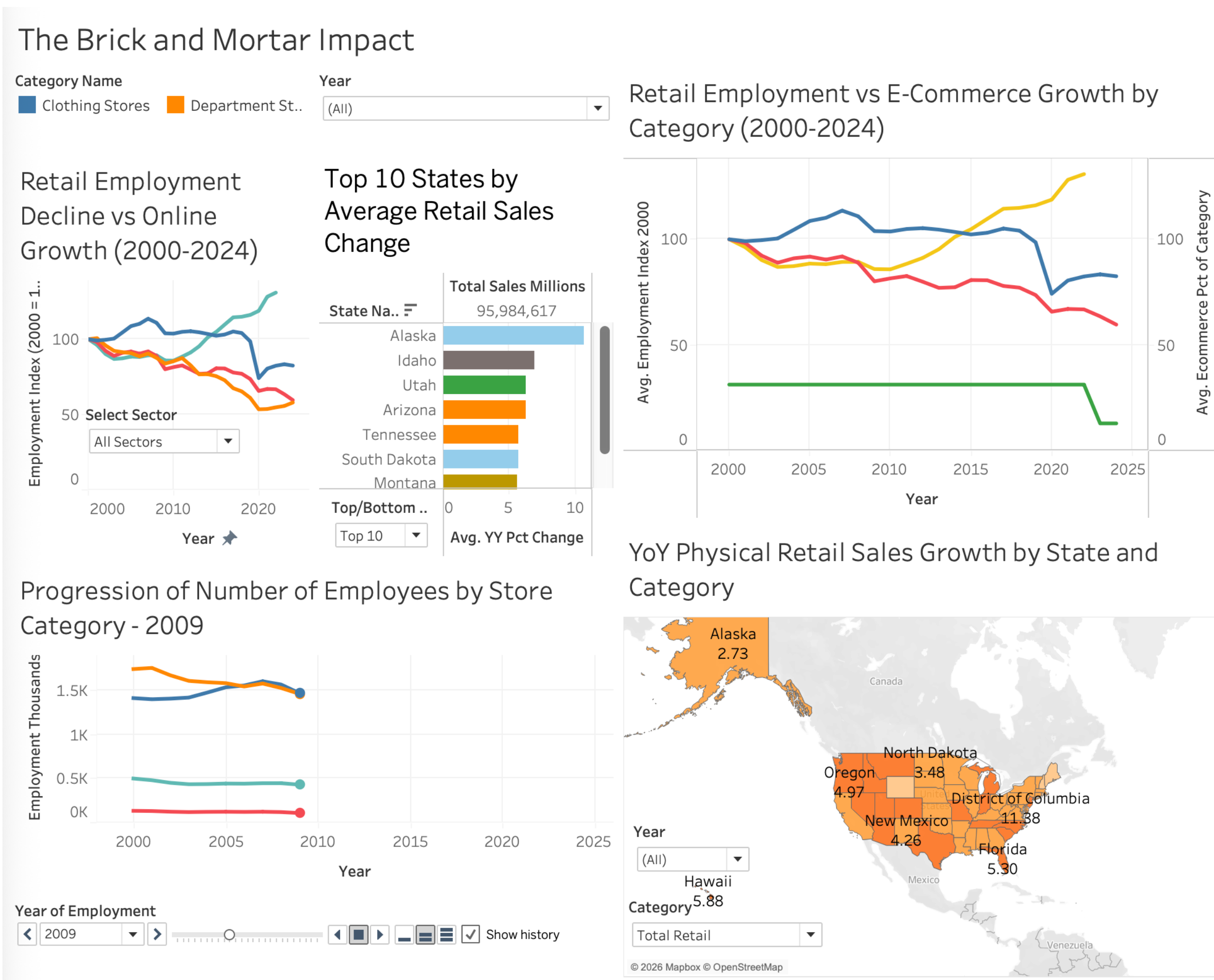
3.7 Dashboard 1: The E-Commerce Rise

The E-Commerce Rise dashboard brings together four visualizations that collectively answer Research Question 1. The quarterly sales area chart anchors the dashboard with the 25-year growth story, giving viewers the macro context before they explore the details. The radial chart shows the average e-commerce share by retail category at a glance, making it immediately clear which sectors have moved most online. The online vs. brick-and-mortar bar chart with the year parameter and drill-down lets viewers explore the sales balance by category for any year between 2018 and 2025, showing how the split has shifted over time. The category size vs. e-commerce share scatter plot rounds out the dashboard by mapping where each category sits in terms of both market size and online adoption, highlighting non-store retailers as a clear outlier and identifying categories like food and beverage where significant digital growth still lies ahead.



3.8 Dashboard 2: The Brick-and-Mortar Impact

The Brick-and-Mortar Impact dashboard addresses Research Question 2 by focusing on the consequences of the e-commerce shift across the workforce and geography. The retail employment decline vs. online growth dual-axis chart shows the direct relationship between rising e-commerce penetration and falling employment in physical retail sectors. The progression of employees by store category animated chart steps through each year from 2000 to 2024, making the divergence between physical and online retail employment visible as it unfolds over time. The retail employment vs. e-commerce growth by category line chart provides a broader view of how each sector's workforce trajectory compares across the full period. The year-over-year physical retail sales map shows which states are experiencing the steepest declines in brick-and-mortar sales, with the category parameter allowing viewers to filter by sector. The top 10 states by retail sales change chart provides a ranked view of the best and worst performing states, offering a concrete geographic takeaway to close the dashboard.



4. Conclusion

4.1 Conclusions for Research Question 1

E-commerce has grown from a negligible share of total U.S. retail in 1999 to over 16% of total retail by 2024, and it is still growing. The shift has not been equal across categories. Non-store retailers already operate almost entirely online, with penetration rates above 66%. Among traditional retail categories, clothing and general merchandise have moved online the most, while motor vehicles and parts remain almost entirely physical. The pace of the shift in each category appears to be driven by how easy it is to standardize and ship the product, how comfortable consumers are with returns, and how aggressively online competitors have entered the space.

4.2 Conclusions for Research Question 2

The trajectory of physical retail varies by category and geography, but the overall direction is contraction. Electronics and department stores have lost 30 to 40% of their 2000 employment levels by 2024. The geographic data show that the shift is furthest along in Northeastern markets and least advanced in Sun Belt states where population growth is likely partly offsetting the broader decline. Categories with low current e-commerce penetration, like food and beverage and building materials, are likely the next to see significant disruption as delivery infrastructure improves.

4.3 Possible Explanations for Observed Trends

The steep decline in electronics retail employment could line up with Amazon's early and aggressive move into that category, where products became easy to standardize, price competitively, and ship reliably. Clothing stores held up through 2019 but dropped sharply after 2020, which points to the pandemic as a turning point for apparel consumers who previously preferred trying items in person. The stronger retail performance in Sun Belt states likely reflects both population growth and it may suggest that high-speed internet and smartphone adoption reached saturation later in those markets, meaning e-commerce is still earlier in its disruption cycle there.

4.4 Future Research Questions

Several questions come out of this analysis that would be worth exploring further:

1. To what extent does the growth of e-commerce fulfillment employment offset the decline in physical retail employment at the local level, and are these new jobs accessible to workers who lost retail positions?

2. How do e-commerce penetration rates and physical retail employment trends differ across income levels and demographic groups, and are lower-income communities bearing a disproportionate share of the job losses?

3. What is the relationship between commercial real estate vacancy rates and e-commerce penetration at the metro level, and can vacancy data serve as a leading indicator of further physical retail contraction?

4. How will the expansion of same-day and next-day delivery in food and beverage and building materials affect e-commerce penetration in those currently low-penetration categories over the next decade?

5. Does the stronger retail performance in Sun Belt states reflect a permanent structural difference, or will those markets eventually converge toward the higher e-commerce penetration seen in Northeastern states as their populations age and digital adoption matures?

Appendix: Data Sources

Source	Publisher	URL
E-Commerce Retail Sales (ECOMSA)	FRED / U.S. Census Bureau	fred.stlouisfed.org/series/ECOMSA
Quarterly E-Commerce Supplemental Tables	U.S. Census Bureau	census.gov/retail/ecommerce/historic_releases.html
Retail Employment by Sector	BLS via FRED	fred.stlouisfed.org
Monthly State Retail Sales	U.S. Census Bureau	census.gov/retail/state_retail_sales.html